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Accelerator Design Educational Primer – Conceptualizing and Optimizing the Hybrid LHeC-like Electron-Ion Collider Design

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The Electron-Ion Collider (EIC) Mission Need requires $\sqrt{s} = 20\text{--}100$ GeV (upgradable to 140 GeV) and luminosity $10^{33}\text{--}10^{34}$ cm $^{-2}$ s $^{-1}$. The current ring-ring baseline achieves the full scope, including $\sim 10^{34}$ cm $^{-2}$ s $^{-1}$ across all energies. However, when the design is re-optimized for the lower boundary —accepting $\sim 10^{33}$ cm $^{-2}$ s $^{-1}$ and prioritizing cost —an alternative configuration emerges as more advantageous: a hybrid LHeC-like electron accelerator using multi-pass energy recovery linacs (ERL).

This solution reduces electron-beam power by roughly an order of magnitude, yielding nearly a factor of two reduction in total project cost compared with the present baseline while still satisfying the minimum physics requirements. The study performs parametric cost and performance modeling, augmented by AI-driven optimization, to explore this design space.

Serving primarily as an educational exercise for the next generation of accelerator physicists and engineers, the paper demonstrates modern design methods: rapid parametric scans, cost-driven optimization, and integration of AI tools. It examines technical feasibility, identifies critical R&D (high-current ERL operation, beam-beam effects, synchronization, etc.), and discusses how such re-optimization studies can be used to train designers in an era when artificial intelligence dramatically expands exploration of complex accelerator parameter spaces.

Footnotes

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In which format do you intend to submit your paper?

LaTeX

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